

Online Learning I: Experts, Linear Classifiers, and Sequences

COMP0078: Supervised Learning

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*Revision notation: items marked with $\star$ are specifically listed in the revision notes file ([revision25-26-V1.pdf](#)).
Non-examinable content (NP-hardness of sparse classifier) has been removed.*

1 Introduction: Batch versus Online Learning

1.1 Batch Learning

- **Model:** Training data set sampled i.i.d. from a distribution.
- **Aim:** Build a classifier from training data that predicts well on new data.
- **Evaluation metric:** Generalization error.

1.2 Online Learning $\star$

- **Model:** Online sequence of data (worst-case model, no distributional assumptions).
- **Aim:** Sequentially predict and update a classifier. No training/test set distinction.
- **Evaluation metric:** Cumulative error (Regret).

2 Part I: Learning with Expert Advice $\star$

2.1 The Model

We assume an online sequence of data $S = (\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m) \in \{0, 1\}^n \times \{0, 1\}$.

Interpretation: The vector $\mathbf{x}_t$ represents the predictions of n experts about an outcome y_t . Expert i predicts $x_{t,i} \in \{0, 1\}$ at time t .

Protocol:

```
for  $t = 1$  to  $m$  do
  Get instance  $\mathbf{x}_t \in \{0, 1\}^n$ 
  Predict  $\hat{y}_t \in \{0, 1\}$ 
  Get label  $y_t \in \{0, 1\}$ 
  Incur loss  $|y_t - \hat{y}_t|$ 
end for
```

Definition 2.1 (Cumulative Loss $\star$). *The loss of algorithm A on sequence S is:*

$$L_A(S) := \sum_{t=1}^m |y_t - \hat{y}_t| \tag{1}$$

The loss of expert i is $L_i(S) := \sum_{t=1}^m |y_t - x_{t,i}|$.

Goal: Bounds of the form $L_A(S) \leq a \min_i L_i(S) + b \log(n)$ where a, b are small constants (Regret bounds).

2.2 The Halving Algorithm*

Assumes the **realizable case** (there exists at least one consistent expert).

Algorithm 1 Halving Algorithm

```

Initialize: All experts are consistent
for  $t = 1$  to  $m$  do
    Predict with majority vote of consistent experts
    Receive  $y_t$ 
    if mistake then
        Remove all experts that predicted incorrectly
    end if
end for

```

Proposition 2.2 (Halving Mistake Bound*). *For any sequence with a consistent expert, the Halving Algorithm (Algorithm 1) makes $\leq \log_2 n$ mistakes.*

Proof. Each mistake removes at least half of the remaining consistent experts. Starting with n experts, we can halve at most $\log_2 n$ times. $\square$

2.3 The Weighted Majority (WM) Algorithm*

Algorithm 2 Weighted Majority Algorithm

```

Initialize:  $w_{1,i} = 1$  for all  $i = 1, \dots, n$ ; parameter  $\beta \in [0, 1)$ 
for  $t = 1$  to  $m$  do
    Predict 1 if  $\sum_{i: x_{t,i}=1} w_{t,i} \geq \sum_{i: x_{t,i}=0} w_{t,i}$ , else 0
    Receive  $y_t$ 
    if  $x_{t,i} \neq y_t$  then
         $w_{t+1,i} = \beta w_{t,i}$ 
    else
         $w_{t+1,i} = w_{t,i}$ 
    end if
end for

```

Notation: M = mistakes of algorithm; M_i = mistakes of expert i ; $w_{t,i} = \beta^{M_{t,i}}$; $W_t = \sum_{i=1}^n w_{t,i}$.

Key observation: If no mistake: $W_{t+1} \leq W_t$. If mistake: $W_{t+1} \leq \frac{1+\beta}{2} W_t$.

Theorem 2.3 (Weighted Majority Bound*). *Let M be the number of mistakes of WM (Algorithm 2), and M_i the mistakes of expert i . Then:*

$$M \leq \frac{\ln(1/\beta)}{\ln(2/(1+\beta))} M_i + \frac{\ln n}{\ln(2/(1+\beta))} \quad (2)$$

Setting $\beta = 1/e$: $M \leq 2.63 M_i + 2.63 \ln n$.

Proof. After M mistakes: $W_{m+1} \leq \left(\frac{1+\beta}{2}\right)^M n$.

For any expert i : $w_{m+1,i} = \beta^{M_i}$. Since $W_{m+1} \geq w_{m+1,i}$:

$$\beta^{M_i} \leq \left(\frac{1+\beta}{2}\right)^M n \quad (3)$$

Taking logarithms:

$$M_i \ln \beta \leq M \ln \left(\frac{1+\beta}{2}\right) + \ln n \quad (4)$$

Rearranging:

$$M \ln \left(\frac{2}{1+\beta}\right) \leq M_i \ln(1/\beta) + \ln n \quad (5)$$

Dividing by $\ln(2/(1+\beta))$ gives the result. $\square$

2.4 Refining the Experts Model: General Loss Functions*

More generally, for arbitrary loss functions $L : Y \times \hat{Y} \rightarrow [0, +\infty]$, we seek:

$$L_A(S) - \min_{i \in [n]} L_i(S) \leq o(m) \quad (6)$$

Definition 2.4 (Entropic Loss*). For $Y = \{0, 1\}$ and $\hat{Y} = [0, 1]$:

$$L_{en}(y, \hat{y}) := y \ln \frac{y}{\hat{y}} + (1-y) \ln \frac{1-y}{1-\hat{y}} \quad (7)$$

Note: For $y \in \{0, 1\}$, if $y = 1$, then $L_{en}(1, \hat{y}) = -\ln \hat{y}$.

2.5 Weighted Average (WA) Algorithm*

Algorithm 3 Weighted Average (WA) Algorithm

Initialize: $\mathbf{v}_1 := \mathbf{w}_1 := (1/n, \dots, 1/n)$

Select: Learning rate $\eta > 0$, Loss function L

for $t = 1$ to m **do**

Receive $\mathbf{x}_t \in [0, 1]^n$

Predict $\hat{y}_t := \mathbf{v}_t \cdot \mathbf{x}_t = \sum_{i=1}^n v_{t,i} x_{t,i}$

Receive $y_t \in [0, 1]$

Incur loss $L_{WA} := L_{WA} + L(y_t, \hat{y}_t)$, $L_i := L_i + L(y_t, x_{t,i})$

Update: $v_{t+1,i} := \frac{v_{t,i} e^{-\eta L(y_t, x_{t,i})}}{\sum_{j=1}^n v_{t,j} e^{-\eta L(y_t, x_{t,j})}}$

end for

Theorem 2.5 (Regret of Weighted Average*). For all sequences $S = (\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m) \in [0, 1]^n \times [0, 1]$, the WA algorithm (Algorithm 3) satisfies:

$$L_{WA}(S) - \min_i L_i(S) \leq \frac{1}{\eta} \ln(n) \quad (8)$$

For entropic loss (Definition 2.4) with $\eta = 1$: bound is $\ln n$. For square loss with $\eta = 1/2$: bound is $2 \ln n$.

Proof of Theorem 2.5.* Let $\Delta_n := \{\mathbf{x} \in [0, 1]^n : \sum_{i=1}^n x_i = 1\}$ and $d(\mathbf{u}, \mathbf{v}) := \sum_{i=1}^n u_i \ln \frac{u_i}{v_i}$ (KL divergence).

Note: $L_{en}(y, \hat{y}) = d((y, 1-y), (\hat{y}, 1-\hat{y}))$.

Key equality (for $\mathbf{u} \in \Delta_n$):

$$L_{en}(y_t, \hat{y}_t) - \sum_{i=1}^n u_i L_{en}(y_t, x_{t,i}) = d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1}) \quad (9)$$

Proof of (9): For $y_t = 1$ (using $L_{en}(1, x) = -\ln x$):

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1}) &= \sum_{i=1}^n u_i \ln \frac{v_{t+1,i}}{v_{t,i}} \\ &= \sum_{i=1}^n u_i \ln \frac{v_{t,i} e^{-L(1, x_{t,i})}}{\sum_j v_{t,j} e^{-L(1, x_{t,j})}} \cdot \frac{1}{v_{t,i}} \\ &= \sum_{i=1}^n u_i (-L(1, x_{t,i}) - \ln(\hat{y}_t)) \\ &= L(1, \hat{y}_t) - \sum_{i=1}^n u_i L(1, x_{t,i}) \end{aligned}$$

Summing over t :

$$\sum_{t=1}^m L_{en}(y_t, \hat{y}_t) - \sum_{t=1}^m \sum_{i=1}^n u_i L_{en}(y_t, x_{t,i}) = d(\mathbf{u}, \mathbf{v}_1) - d(\mathbf{u}, \mathbf{v}_{m+1}) \quad (10)$$

Since $d(\mathbf{u}, \mathbf{v}_1) \leq \ln n$ and $d(\mathbf{u}, \mathbf{v}_{m+1}) \geq 0$, choosing $\mathbf{u}$ as the unit vector for the best expert gives the result. $\square$

2.6 Hedge Algorithm*

Allocation setting: Learner plays allocation $\mathbf{v}_t \in \Delta_n$, nature returns loss vector $\ell_t \in [0, 1]^n$.

Two models:

1. **HA-1:** Incur loss $L_A(t) := \mathbf{v}_t \cdot \ell_t$
2. **HA-2:** Randomly select action $\hat{y} \in [n]$ with $\text{Prob}(j) = v_{t,j}$. Then $\mathbb{E}[L_{HA}] = \mathbf{v}_t \cdot \ell_t$

Algorithm 4 Hedge Algorithm

Initialize: $\mathbf{v}_1 := (1/n, \dots, 1/n)$; Select $\eta > 0$

for $t = 1$ **to** m **do**

 Predict $\mathbf{v}_t$ (or $\hat{y}_t \sim \mathbf{v}_t$ for HA-2)

 Receive loss $\ell_t \in [0, 1]^n$

 Incur loss, update: $v_{t+1,i} := \frac{v_{t,i} e^{-\eta \ell_{t,i}}}{\sum_{j=1}^n v_{t,j} e^{-\eta \ell_{t,j}}}$

end for

Theorem 2.6 (Hedge Bound*). *For all sequences $S = \ell_1, \dots, \ell_m \in [0, 1]^n$ with $\eta = \sqrt{2 \ln n / m}$, the Hedge algorithm (Algorithm 4) satisfies:*

$$\mathbb{E}[L_{HA}(S)] - \min_i L_i(S) \leq \sqrt{2m \ln n} \quad (11)$$

Proof of Theorem 2.6.* Let $Z_t := \sum_{i=1}^n v_{t,i} \exp(-\eta \ell_{t,i})$. For any $\mathbf{u} \in \Delta_n$:

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1}) &= \sum_{i=1}^n u_i \ln \frac{v_{t+1,i}}{v_{t,i}} \\ &= -\eta \mathbf{u} \cdot \boldsymbol{\ell}_t - \ln \sum_{i=1}^n v_{t,i} e^{-\eta \ell_{t,i}} \\ &\geq -\eta \mathbf{u} \cdot \boldsymbol{\ell}_t - \ln \sum_{i=1}^n v_{t,i} \left(1 - \eta \ell_{t,i} + \frac{\eta^2 \ell_{t,i}^2}{2} \right) \end{aligned} \quad (12)$$

$$\begin{aligned} &= -\eta \mathbf{u} \cdot \boldsymbol{\ell}_t - \ln \left(1 - \eta \mathbf{v}_t \cdot \boldsymbol{\ell}_t + \frac{\eta^2}{2} \sum_{i=1}^n v_{t,i} \ell_{t,i}^2 \right) \\ &\geq \eta(\mathbf{v}_t \cdot \boldsymbol{\ell}_t - \mathbf{u} \cdot \boldsymbol{\ell}_t) - \frac{\eta^2}{2} \sum_{i=1}^n v_{t,i} \ell_{t,i}^2 \end{aligned} \quad (13)$$

using $e^{-x} \leq 1 - x + x^2/2$ for (12) and $\ln(1+x) \leq x$ for (13).

Summing over t :

$$\sum_{t=1}^m (\mathbf{v}_t \cdot \boldsymbol{\ell}_t - \mathbf{u} \cdot \boldsymbol{\ell}_t) \leq \frac{\ln n}{\eta} + \frac{\eta}{2} \sum_{t=1}^m \sum_{i=1}^n v_{t,i} \ell_{t,i}^2 \leq \frac{\ln n}{\eta} + \frac{\eta m}{2} \quad (14)$$

For unit vector $\mathbf{u}$ (best expert) and $\eta = \sqrt{2 \ln n / m}$, the result follows. $\square$

3 Part II: Online Learning of Linear Classifiers*

3.1 Setup

Sequence: $S = (\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)$

Comparison class: $U = \{\mathbf{u}\}$ (hypothesis space)

Goal: Bound $L_A(S) - \inf_{\mathbf{u} \in U} \text{Loss}_{\mathbf{u}}(S)$ for arbitrary sequence S .

We focus on the **realizable case**: $\exists \mathbf{u} \in U$ s.t. $\text{Loss}_{\mathbf{u}}(S) = 0$.

3.2 The Perceptron*

Assumption: Data is linearly separable by margin γ . Exists $\mathbf{v}$ with:

1. $\|\mathbf{v}\| = 1$
2. $\forall t : y_t \in \{-1, +1\}, \|\mathbf{x}_t\| \leq R$
3. $\forall t : y_t(\mathbf{x}_t \cdot \mathbf{v}) \geq \gamma$

Lemma 3.1 (Pythagorean Lemma*). *If $(\mathbf{w}_t \cdot \mathbf{x}_t)y_t < 0$ then $\|\mathbf{w}_{t+1}\|^2 \leq \|\mathbf{w}_t\|^2 + \|\mathbf{x}_t\|^2$.*

Proof. $\|\mathbf{w}_{t+1}\|^2 = \|\mathbf{w}_t + y_t \mathbf{x}_t\|^2 = \|\mathbf{w}_t\|^2 + 2(\mathbf{w}_t \cdot \mathbf{x}_t)y_t + \|\mathbf{x}_t\|^2 \leq \|\mathbf{w}_t\|^2 + \|\mathbf{x}_t\|^2$. $\square$

Lemma 3.2 (Upper Bound on Weight Norm*). $\|\mathbf{w}_t\|^2 \leq M_t R^2$.

Proof. By induction using Lemma 3.1. Base: $M_1 = 0, \|\mathbf{w}_1\|^2 = 0$. Induction step: if a mistake occurs on trial t , then $\|\mathbf{w}_{t+1}\|^2 \leq \|\mathbf{w}_t\|^2 + \|\mathbf{x}_t\|^2 \leq M_t R^2 + R^2 = M_{t+1} R^2$ (by Lemma 3.1). $\square$

Lemma 3.3 (Lower Bound on Weight Norm*). $M_t \gamma \leq \|\mathbf{w}_t\|$.

Algorithm 5 Perceptron Algorithm

Input: $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m) \in \mathbb{R}^n \times \{-1, 1\}$
Initialize $\mathbf{w}_1 = \mathbf{0}$; $M_1 = 0$
for $t = 1$ to m **do**
 Receive $\mathbf{x}_t \in \mathbb{R}^n$
 Predict $\hat{y}_t = \text{sign}(\mathbf{w}_t \cdot \mathbf{x}_t)$
 Receive y_t
 if mistake ($\hat{y}_t y_t \leq 0$) **then**
 $\mathbf{w}_{t+1} = \mathbf{w}_t + y_t \mathbf{x}_t$; $M_{t+1} = M_t + 1$
 else
 $\mathbf{w}_{t+1} = \mathbf{w}_t$; $M_{t+1} = M_t$
 end if
end for

Proof. By Cauchy-Schwarz: $\|\mathbf{w}_t\| \geq \mathbf{w}_t \cdot \mathbf{v}$.

By induction: $\mathbf{w}_{t+1} \cdot \mathbf{v} = (\mathbf{w}_t + y_t \mathbf{x}_t) \cdot \mathbf{v} \geq \mathbf{w}_t \cdot \mathbf{v} + \gamma \geq M_t \gamma + \gamma = M_{t+1} \gamma$. $\square$

Theorem 3.4 (Perceptron Bound (Novikoff)*). *For sequences $S = (\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m) \in \mathbb{R}^n \times \{-1, 1\}$, the Perceptron algorithm (Algorithm 5) satisfies:*

$$M \leq \left(\frac{R}{\gamma} \right)^2 \quad (15)$$

where $R := \max_t \|\mathbf{x}_t\|$ when $\exists \mathbf{v}$ with $\|\mathbf{v}\| = 1$ and $(\mathbf{v} \cdot \mathbf{x}_t) y_t \geq \gamma$ for all t .

Proof of Theorem 3.4.* Combining Lemma 3.3 and Lemma 3.2: $(M\gamma)^2 \leq \|\mathbf{w}_{m+1}\|^2 \leq MR^2 \implies M \leq R^2/\gamma^2$. $\square$

3.3 Online Gradient Descent (OGD)*

Definition 3.5 (Hinge Loss*). $L_{hi}(y, \hat{y}) := \max(0, 1 - y\hat{y})$.

Algorithm 6 Online Gradient Descent

Initialize $\mathbf{w}_1 := \mathbf{0}$; Select $\eta > 0$
for $t = 1$ to m **do**
 Receive $\mathbf{x}_t \in \mathbb{R}^n$
 Predict $\hat{y}_t := \mathbf{w}_t \cdot \mathbf{x}_t$
 Receive $y_t \in \{-1, 1\}$
 Update: $\mathbf{w}_{t+1} := \mathbf{w}_t + [y_t \hat{y}_t < 1] \eta y_t \mathbf{x}_t$
end for

Theorem 3.6 (OGD Regret Bound*). *Given $R = \max_t \|\mathbf{x}_t\|$, $\|\mathbf{u}\| \leq U$ and $\eta := \frac{U}{R\sqrt{m}}$, Algorithm 6 with hinge loss (Definition 3.5) satisfies:*

$$\sum_{t=1}^m L_{hi}(y_t, \hat{y}_t) - L_{hi}(y_t, \mathbf{u} \cdot \mathbf{x}_t) \leq \sqrt{U^2 R^2 m} \quad (16)$$

for any vector $\mathbf{u}$.

Proof of Theorem 3.6.* By convexity of hinge loss: $L_{hi}(y_t, \hat{y}_t) - L_{hi}(y_t, \mathbf{u} \cdot \mathbf{x}_t) \leq (\mathbf{w}_t - \mathbf{u}) \cdot \mathbf{z}_t$ where $\mathbf{z}_t := -y_t \mathbf{x}_t [y_t (\mathbf{w}_t \cdot \mathbf{x}_t) < 1]$.

From update: $\|\mathbf{w}_{t+1} - \mathbf{u}\|^2 = \|\mathbf{w}_t - \eta \mathbf{z}_t - \mathbf{u}\|^2 = \|\mathbf{w}_t - \mathbf{u}\|^2 - 2\eta \langle \mathbf{w}_t - \mathbf{u}, \mathbf{z}_t \rangle + \eta^2 \|\mathbf{z}_t\|^2$.

Thus: $\langle \mathbf{w}_t - \mathbf{u}, \mathbf{z}_t \rangle = \frac{1}{2\eta} (\|\mathbf{w}_t - \mathbf{u}\|^2 - \|\mathbf{w}_{t+1} - \mathbf{u}\|^2 + \eta^2 \|\mathbf{z}_t\|^2)$.

Summing:

$$\sum_{t=1}^m \langle \mathbf{w}_t - \mathbf{u}, \mathbf{z}_t \rangle \leq \frac{1}{2\eta} U^2 + \frac{\eta}{2} m R^2 = \sqrt{U^2 R^2 m} \quad (17)$$

□

3.4 Boolean Functions and Linear Thresholds*

Terminology:

- **Literal:** Single (negated) variable (e.g., $x_1, \bar{x}_5$)
- **Term/Conjunction:** Iterated “and” of literals (e.g., $x_1 x_3$)
- **Clause/Disjunction:** Iterated “or” of literals (e.g., $x_1 \vee x_3$)
- **Monotone:** No negated literals

A linear threshold function $f_{\mathbf{u},b}(\mathbf{x}) := \text{sign}(\mathbf{u} \cdot \mathbf{x} + b)$ can represent disjunctions and conjunctions.

Feature map for non-monotone: $\phi(\mathbf{x}) := (x_1, 1 - x_1, \dots, x_n, 1 - x_n)$.

3.5 Perceptron for Disjunctions*

Corollary 3.7 (Perceptron Disjunction Bound*). *With feature map $\phi(\mathbf{x}) := (\mathbf{x}, 1)$, the Perceptron (Algorithm 5) learns monotone disjunctions with:*

$$M \leq (4k + 1)(n + 1) \quad (18)$$

where k is the number of literals.

Proof. Using the $M \leq R^2 \|\mathbf{u}\|^2$ form of Theorem 3.4. For $\mathbf{x} \in \{0, 1\}^n$ and $\phi(\mathbf{x}) := (\mathbf{x}, 1)$, define:

$$u_i^* := \begin{cases} 2 & i \text{ is a literal} \\ 0 & i \text{ is not a literal} \\ -1 & i \text{ is the bias} \end{cases} \quad (19)$$

Then $\mathbf{u}^*$ separates with margin 1, $\|\phi(\mathbf{x})\|^2 \leq n + 1$, and $\|\mathbf{u}^*\|^2 = 4k + 1$. □

3.6 Winnow Algorithm*

Theorem 3.8 (Winnow Mistake Bound*). *If there exists a consistent k -literal monotone disjunction, the Winnow algorithm (Algorithm 7) satisfies:*

$$M \leq 3k(\log n + 1) + 2 \quad (20)$$

Proof. Let $W_t = \sum_{i=1}^n w_{t,i}$.

Positive mistakes ($y_t = 1$): Each mistake doubles at least one relevant weight. Relevant weights never decrease. Once $w_{t,i} \geq n$ it doesn't change. Thus $M_p \leq k(\log n + 1)$.

Negative mistakes ($y_t = 0$): $W_1 = n$. On positive mistake: $W_{t+1} \leq W_t + n$. On negative mistake: $W_{t+1} \leq W_t - n/2$. Thus: $W_{m+1} \leq n + M_p n - M_f n/2$, giving $M_f \leq 2k(\log n + 1) + 2$.

Therefore: $M = M_p + M_f \leq 3k(\log n + 1) + 2$. □

Remark 3.9 (Perceptron vs. Winnow*). *For disjunction learning, the Winnow bound (Theorem 3.8) is $O(k \log n)$, an exponential improvement over the Perceptron bound (Corollary 3.7) of $O(kn)$ with respect to the dimension n . However, for general linear threshold learning the bounds are incomparable: Winnow (like Lasso) prefers sparse hypotheses.*

Algorithm 7 Winnow Algorithm

Input: $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m) \in \{0, 1\}^n \times \{0, 1\}$

Initialize $\mathbf{w}_1 = \mathbf{1}$

for $t = 1$ to m **do**

Receive $\mathbf{x}_t \in \{0, 1\}^n$

Predict: $\hat{y}_t = \begin{cases} 0 & \mathbf{w}_t \cdot \mathbf{x}_t < n \\ 1 & \mathbf{w}_t \cdot \mathbf{x}_t \geq n \end{cases}$

Receive $y_t \in \{0, 1\}$

if mistake **then**

$w_{t+1,i} = w_{t,i} 2^{(y_t - \hat{y}_t)x_{t,i}}$ for $1 \leq i \leq n$

end if

end for

3.7 DNF Learning*

Definition 3.10 (DNF*). *Disjunctive Normal Form is a disjunction of terms (e.g., $x_1x_4x_7 \vee x_1\bar{x}_2 \vee x_2x_5$).*

Feature expansion for k -term DNF (Definition 3.10):

$$\Phi(\mathbf{x}) \cdot \Phi(\mathbf{y}) = \prod_{i=1}^n (1 + x_i y_i) = K_{anova}(\mathbf{x}, \mathbf{y}) \quad (21)$$

where $\Phi(\mathbf{x})$ maps to all monomials. Computable in $O(n)$ time.

Perceptron (Algorithm 5): $\mathbf{w}_t = \sum_{q \in \text{mistakes}} \alpha_q \Phi(\mathbf{x}_q)$.

- Prediction time: $O(n \cdot M)$

- Mistake bound: $O(k2^n)$

Winnow (Algorithm 7): $w_{t,i} = \exp(-\eta \sum_{q \in \text{mistakes}} \alpha_q \Phi(\mathbf{x}_q)_i)$.

- Prediction time: $\Omega(2^n M)$

- Mistake bound: $O(kn)$

4 Part III: Learning with Sequences of Experts*

4.1 Shifting Experts

Setup: For sequence $S = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_\ell, y_\ell)\}$, define loss relative to a sequence of experts:

$$\text{Loss}_{\{i_1, \dots, i_\ell\}}(S) = \sum_{t=1}^{\ell} (x_{t,i_t} - y_t)^2 \quad (22)$$

Goal: Bound $L_A(S)$ relative to best sequence with k shifts, where:

$$k := \sum_{t=1}^{\ell-1} [i_t \neq i_{t+1}] \quad (23)$$

4.2 Fixed Share Algorithm*

Parameters: $\eta > 0, \alpha \in [0, 1]$

Initialization: $w_{1,1}^s = \dots = w_{1,n}^s = 1/n$

Prediction: $v_{t,i} = w_{t,i}^s / W_t$ where $W_t = \sum_i w_{t,i}^s$. Predict $\hat{y}_t = \mathbf{v}_t \cdot \mathbf{x}_t$.

Loss Update: $w_{t,i}^m = w_{t,i}^s e^{-\eta L(y_t, x_{t,i})}$

Share Update:

$$\text{pool} = \sum_{i=1}^n \alpha w_{t,i}^m \quad (24)$$

$$w_{t+1,i}^s = (1 - \alpha) w_{t,i}^m + \frac{1}{n-1} (\text{pool} - \alpha w_{t,i}^m) \quad (25)$$

4.3 Variable Share Algorithm

Same as Fixed Share (Section 4.2) except:

$$\text{pool} = \sum_{i=1}^n (1 - (1 - \alpha)^{L(y_t, x_{t,i})}) w_{t,i}^m \quad (26)$$

$$w_{t+1,i}^s = (1 - \alpha)^{L(y_t, x_{t,i})} w_{t,i}^m + \frac{1}{n-1} (\text{pool} - (1 - (1 - \alpha)^{L(y_t, x_{t,i})}) w_{t,i}^m) \quad (27)$$

4.4 Shifting Loss Bounds*

Theorem 4.1 (Shifting Experts Bound*). *Let $L^* := \text{Loss}_{\{i_1, \dots, i_\ell\}}(S)$ and choose $\alpha \in [0, 1]$. Then the Fixed Share algorithm (Section 4.2) satisfies:*

$$L_A(S) \leq L^* + c[(\ell - 1)(H(\alpha^*) + D(\alpha^* \| \alpha)) + k \ln(n - 1) + \ln n] \quad (28)$$

where $\alpha^* = k/(\ell - 1)$, $H(p) = \frac{1}{p} \ln \frac{1}{p} + \frac{1}{1-p} \ln \frac{1}{1-p}$, and $D(p^* \| p) = p^* \ln \frac{p^*}{p} + (1 - p^*) \ln \frac{1-p^*}{1-p}$.

Setting $\alpha = k/(\ell - 1)$:

$$L_A(S) \leq L^* + c \left[k \ln \frac{\ell - 1}{k} + k \ln(n - 1) + \ln n + k \right] \quad (29)$$

*Proof Sketch**. Define meta-experts with initial weights:

$$w_{1,\{i_1, \dots, i_\ell\}} := \frac{1}{n} (1 - \alpha)^{\ell-1-k} \left(\frac{\alpha}{n-1} \right)^k \quad (30)$$

These sum to 1 and describe all possible expert sequences. The WA algorithm bound (Theorem 2.5) with these meta-experts gives:

$$L_A(S) \leq L^* + c \ln \frac{1}{w_{1,\{i_1, \dots, i_\ell\}}} \quad (31)$$

Computing $\ln w_{1,\{i_1, \dots, i_\ell\}}$ and optimizing over α yields the result. Fixed-share simulates this meta-expert algorithm in $O(n)$ time. $\square$